

Chapter 7: Variational Autoencoder

Variational Inference: Understand the limitation of the Naïve Monte Carlo.

Variational Inference: Understand the importance sampling estimator and the formulation of the variational inference.

Variational Autoencoder: Show that the ELBO is a lower bound of the log-likelihood and becomes a tight bound if and only if we have the perfect proposal $q_{\phi}(\mathbf{z}|\mathbf{x}) = p_{\theta}(\mathbf{z}|\mathbf{x})$.

Variational Autoencoder: Understand the role of reconstruction probability and the KL-penalty in ELBO and how to compute them.

Multi-modal VAE: Understand the model density of Multi-modal VAE.

Multi-modal VAE: Understand how the Multi-modal VAE addresses missing modalities.

Temporal VAE: Understand the model density of Stochastic Variational Video Prediction.

Temporal VAE: Understand how the Stochastic Variational Video Prediction sequentially generates videos.